BRENDA RAMIREZ

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EDUCATION

University of California, San Diego Graduation: June 2027

B.S. Computer Science, with Minor in Design

- *Relevant Coursework: Systems programming and Software Tools,* Data Structures & Object Oriented Design, Computer Organization, Algorithms and Analysis, Foundations of Programming, Machine Learning

EXPERIENCE

Software Engineering Intern San Diego, CA

UC San Diego Computer Science Department | **HTML, CSS, Figma, Python** June 2025 - September 2025

- Advanced a **research initiative**, transforming **data analysis** into a **web-based planning tool** for CCC-to-UC Computer Science transfer pathways
- Built a **Python-based backend** generating **6,000+ optimized transfer plans** across **12 CCC**, applying prerequisite, GE, and major completion rules
- Designed **algorithms** to analyze **articulation agreement conflicts** and measure the academic burden across UC campuses
- **Designed** and **Prototyped** a **student-centered interface** using **HTML/CSS** and **Figma**, improving **accessibility** and **usability** for transfer planning
- Conducted **user testing sessions** and created **iterative UI prototypes** to test transfer-planning workflows before backend integration
- Collaborated in a **diversity-focused computing cohort**, fostering inclusive design practices and **mentorship** in tech research

Undergrad Researcher San Diego, CA

UC San Diego Computer Science Department | **Python, Matplotlib, Seaborn, GitHub** *September 2024 - May 2025*

- Collected and processed **26M+ UC transfer pathways combinations** using **Python** to identify systemic bottlenecks in computer science articulation agreements
- Developed **data-driven visual prototypes** to communicate system bottlenecks and design tradeoffs to both technical and non-technical stakeholders
- Collaborate via **GitHub** for version control, ensuring **reproducibility** and consistency in multi-contributor workflows
- Co-authored a **research paper** and presented findings at a research symposium, addressing bottlenecks in articulation agreements from California Community Colleges (CCC) to the University of California (UC) schools

PROJECTS

Thera Nova - Refugee Health Platform October 2025 - Ongoing

Figma, UI/UX Prototyping, Human-Centered Design, Accessibility, Research

- Designed and **iterated on interactive Figma prototypes** to redesign a nonprofit clinic website, improving multilingual accessibility and user flow for refugee communities
- Applied **human-centered design methods** (interviews, surveys, UX audits, user testing) to identify language and navigation barriers in healthcare access
- Collaborated with nonprofit stakeholders to align **prototype designs with real-world constraints,** accessibility needs, and future technical integrations (including AI-assisted translation exploration)

Portfolio Website *Personal Developer Portfolio* June 2024 - Present

HTML, CSS, JavaScript, Responsive Design, Accessibility

- Developed a **responsive portfolio website** using **HTML** and **CSS Grid/Flexbox** to **showcase technical projects** and coursework
- Implemented a **JavaScript-based theme toggle**, smooth page navigation, and accessibility features for improved **usability** across services
- Structured the site into **modular components** with **reusable layouts** for easy scalability and easy future **project additions**
- Optimized **page load performance** and prepared **deployment setup** for **GitHub Pages** and **Vercel hosting**

UTF-8 Encoding & String Manipulation in C September 2024 - October 2024

C, Memory Management, Unit Testing

- Built a **UTF-8 text analyzer in C** to **validate ASCII**, **capitalize characters**, and **compute string lengths** for multi-byte encodings
- Implemented **substring extraction**, **code point to byte index conversion**, and **multi-byte sequence handling** to support Unicode text processing
- Verified correctness through **comprehensive unit tests across ASCII, multi-byte, and malformed UTF-8 sequences**

SKILLS & INTERESTS

Programming Languages : Java, Python, C++, JavaScript, MATLAB, HTML, CSS
Libraries & Frameworks : Pandas, Numpy, Matplotlib, JUnit, Pytorch, TensorFlow, Seaborn, Pytest
Tools & Platforms : GitHub, Figma (Prototyping & Webframing), Jupyter Notebook, Visual Studio Code, Google Colab
Technical Skills : Full-Stack Development, Data Analysis, Visualization, UI/UX Prototyping, Bilingual, Problem-Solving
Interests : Piano, Thrifting, Baking, Swimming, Play video games, Drawing, Painting